DAQ Core

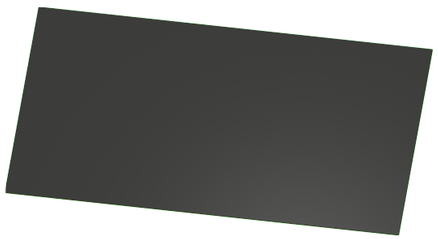
Variant: DRAFT

2026-10-30

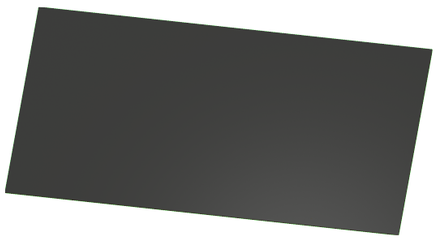
Rev + (Unreleased)

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2	Block Diagram	12	Power - Sequencing	22	.....	32	.....
3	Project Architecture	13	Revision History	23	.....	33	.....
4	MCU	14	.....	24	.....	34	.....
5	MCU Expander	15	.....	25	.....	35	.....
6	MCU IOT	16	.....	26	.....	36	.....
7	In&Out - ADC	17	.....	27	.....	37	.....
8	In&Out - Command	18	.....	28	.....	38	.....
9	Serial Interface	19	.....	29	.....	39	.....
10	Sensor	20	.....	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 30-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

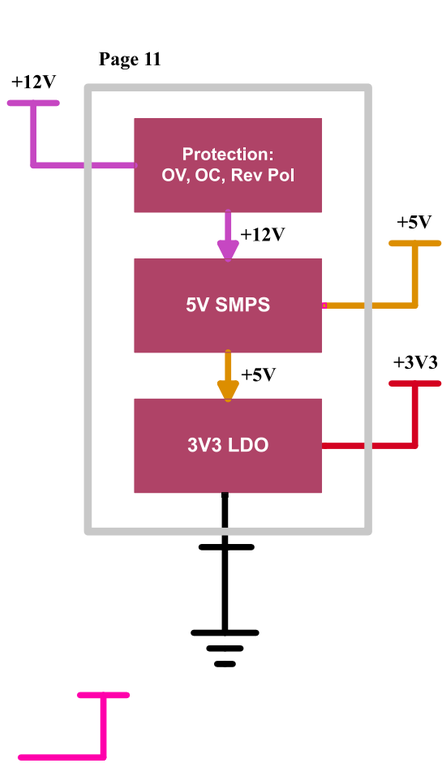
DESIGN NOTE: Example text for cautionary design notes.

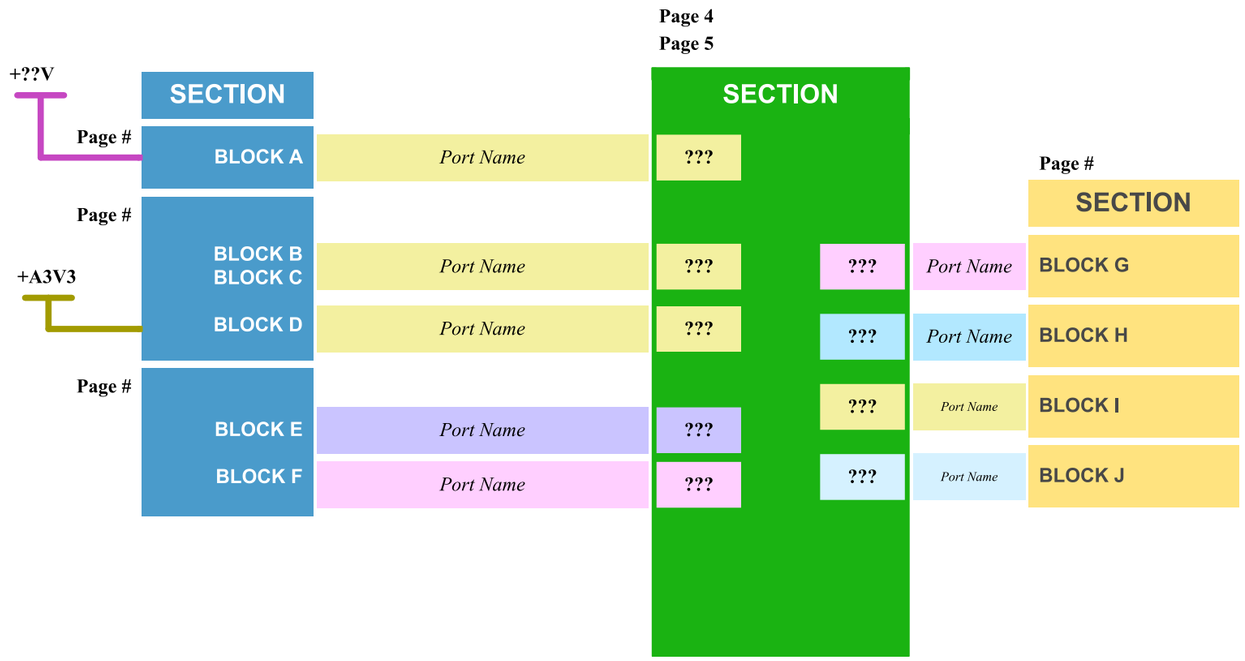
DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 41b59e0
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title:	File Name: Data-Acquisition-Board.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 13

[2] Block Diagram



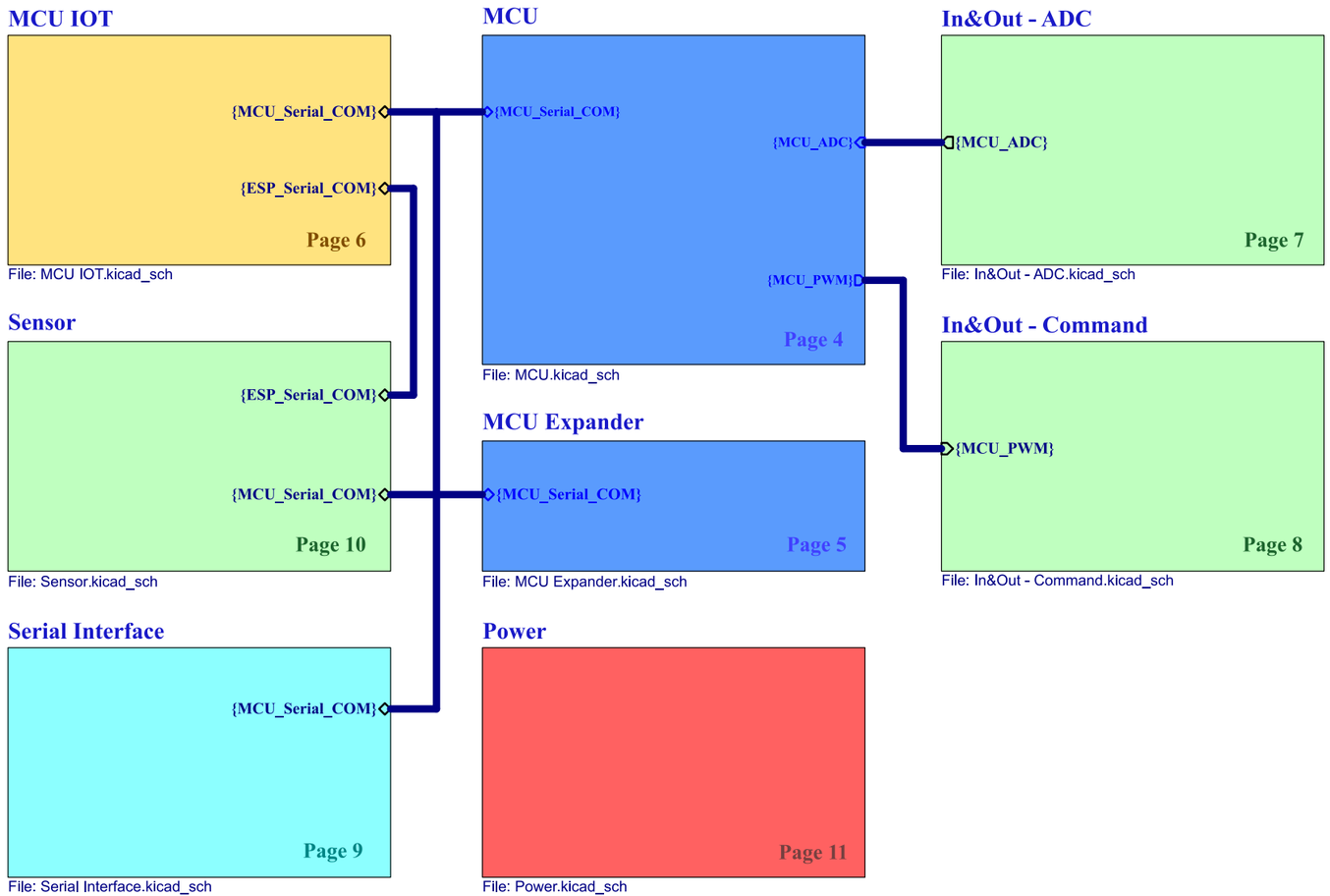


Target specifications:

Input voltage:	7V - 17V
Input power:	7400mW
Spec 3	??
Spec 4	??

Comments:	Company: **Nexvantis**		Variant: DRAFT	Git Hash: 41b59e0
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: **A3**	Sheet: **2** of **13**

[3] Project Architecture



LAYOUT NOTE:
blablabla

DESIGN NOTE:
blablabla

DESIGN NOTE:
blabla

DESIGN NOTE:
blabla

LAYOUT NOTE:
blabla

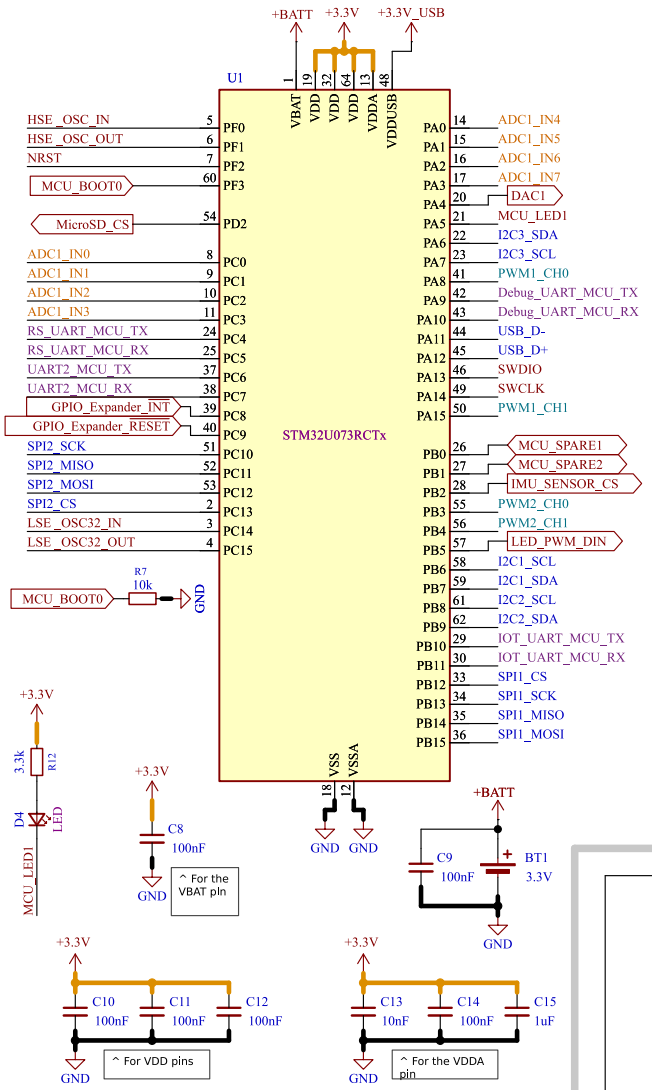
Block Description

Description

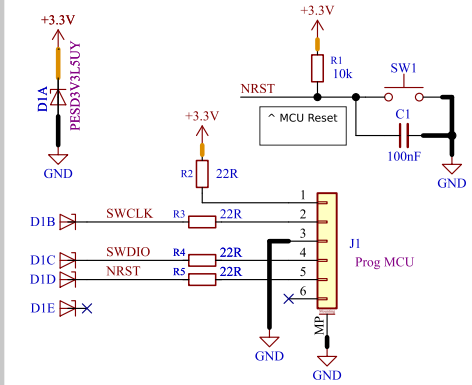
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	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: **A3**	Sheet: **3** of **13**

[4] MCU

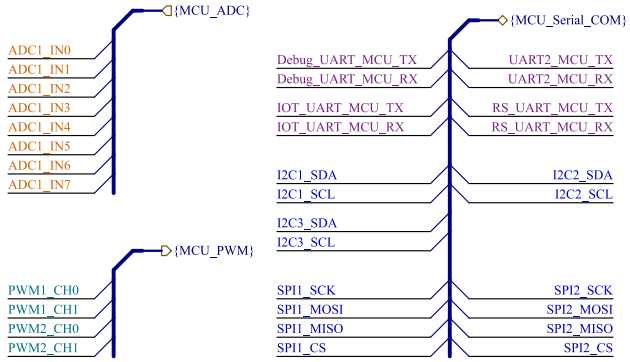
STM32U073RCTx



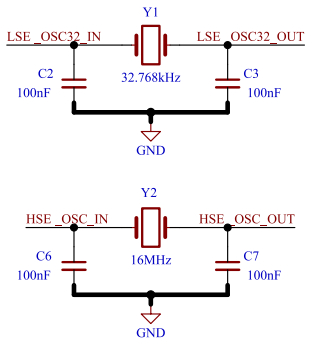
Programming



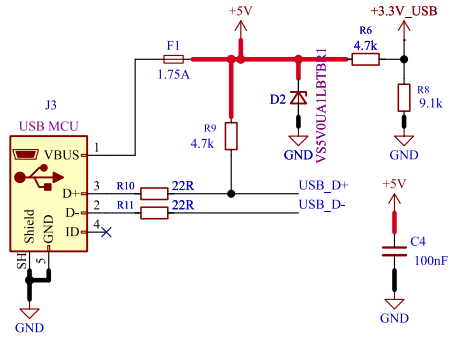
BUS



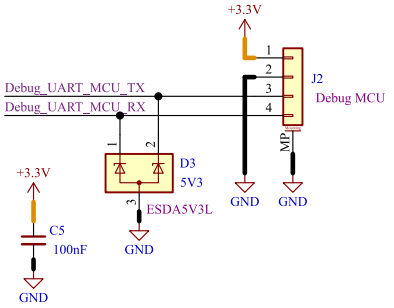
Quartz



USB



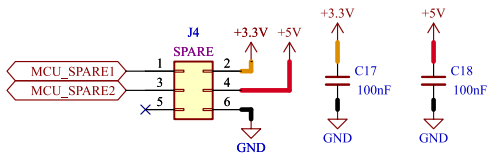
Debug



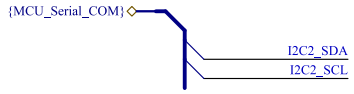
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 41b59e0
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: MCU	File Name: MCU.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU/		Reviewer:	Size: **A4**	Sheet: **4** of **13**

[5] MCU Expander

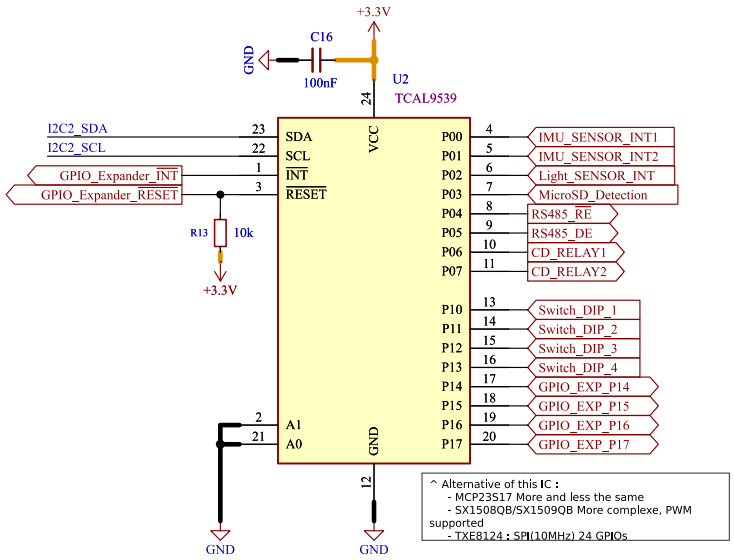
MCU SPARE CONNECTOR



BUS

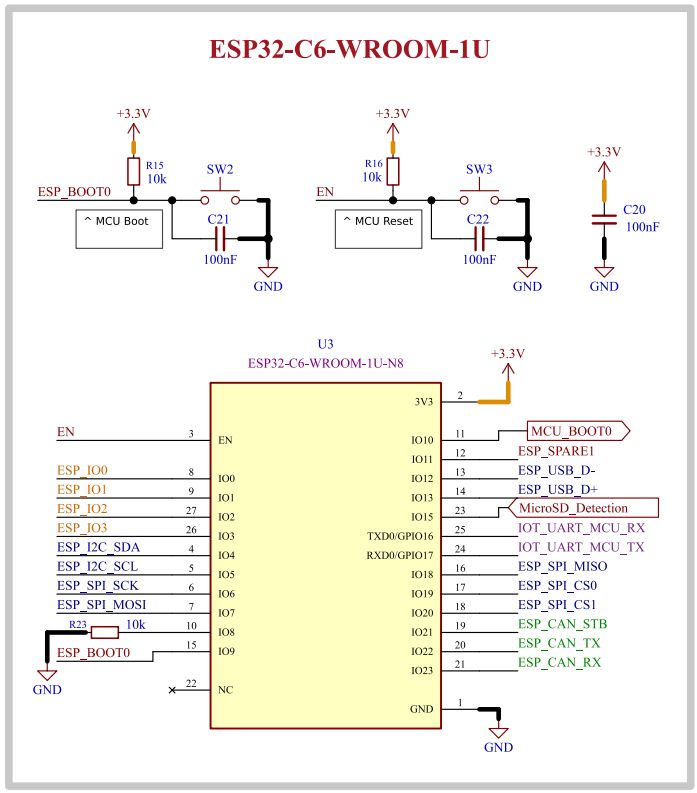


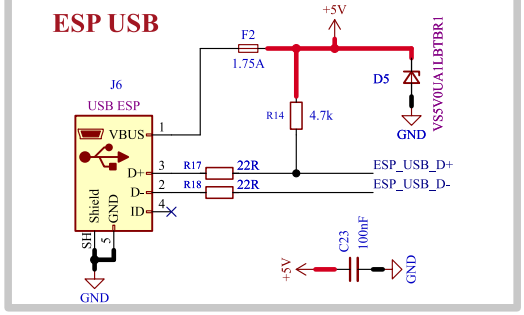
GPIO Expander

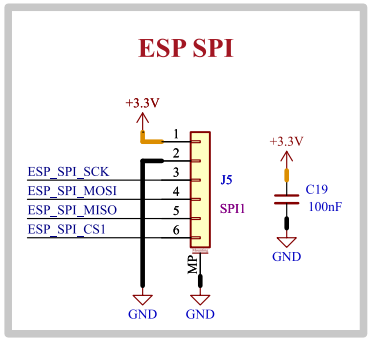


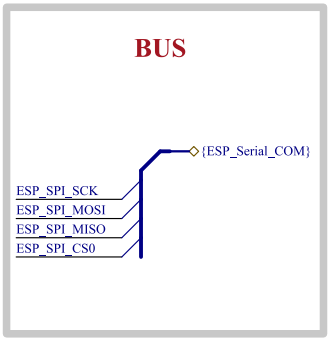
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	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: MCU Expander	File Name: MCU Expander.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
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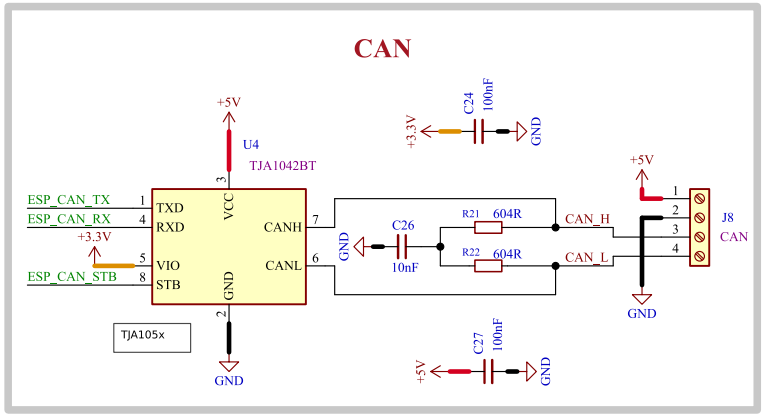
[6] MCU IOT

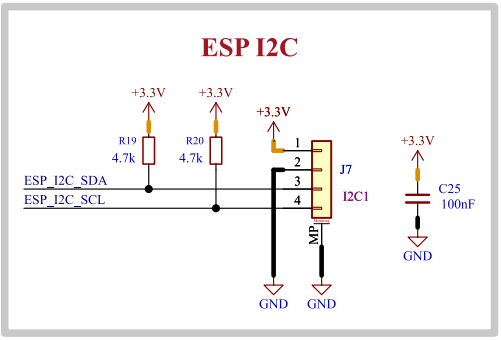


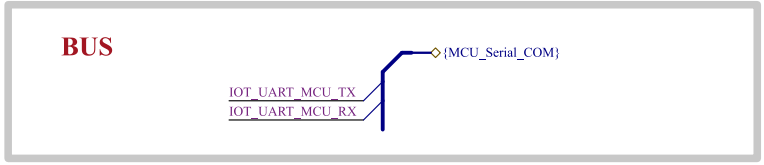


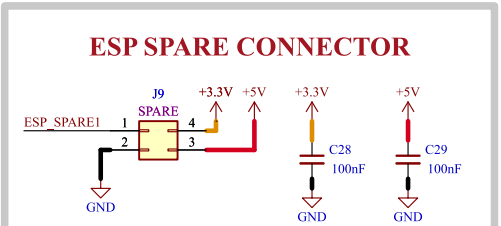


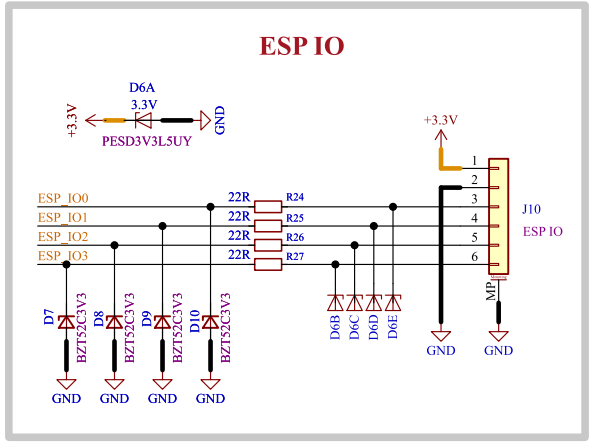








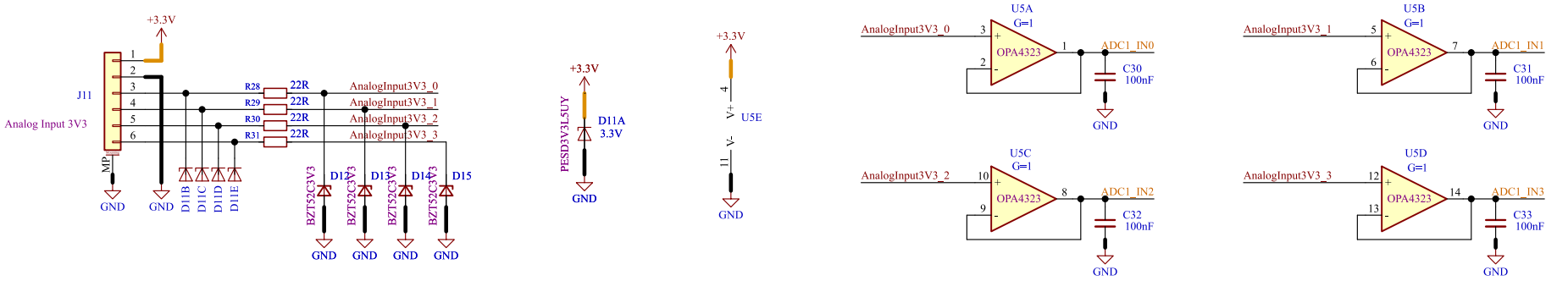




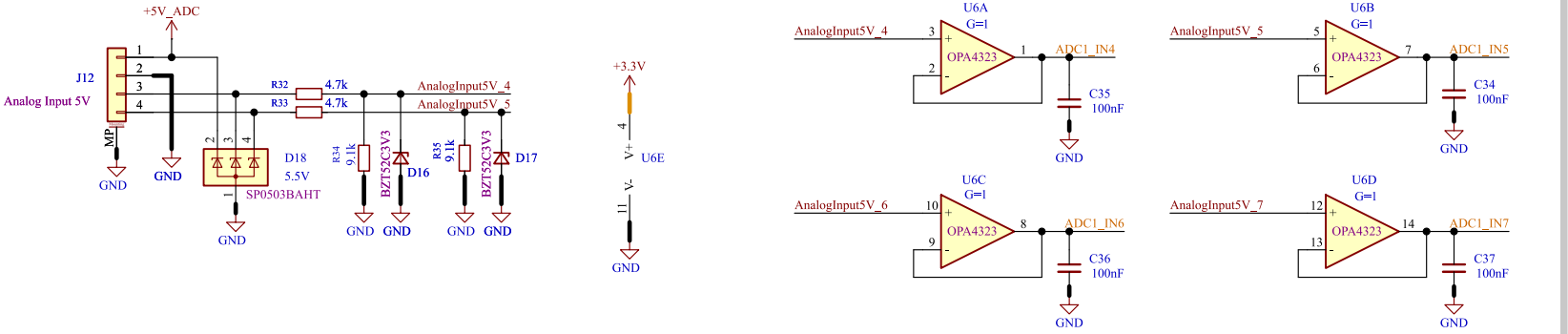
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	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: MCU IOT	File Name: MCU IOT.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/MCU IOT/		Reviewer:	Size: **A4**	Sheet: **6** of **13**

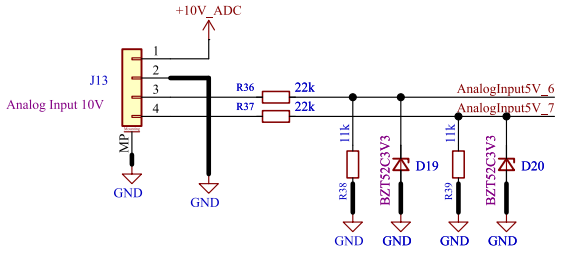
[7] Input&Output ADC

Analog Input ADC 3V3

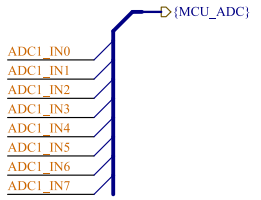


Analog Input ADC 5V & 10V



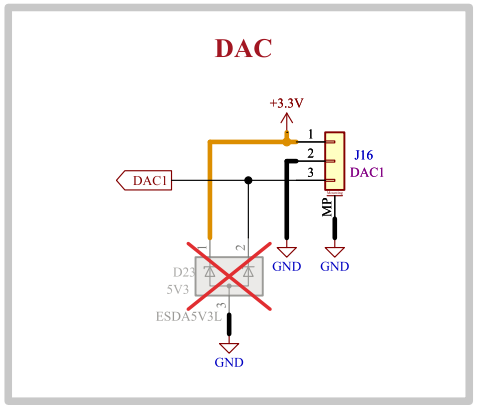


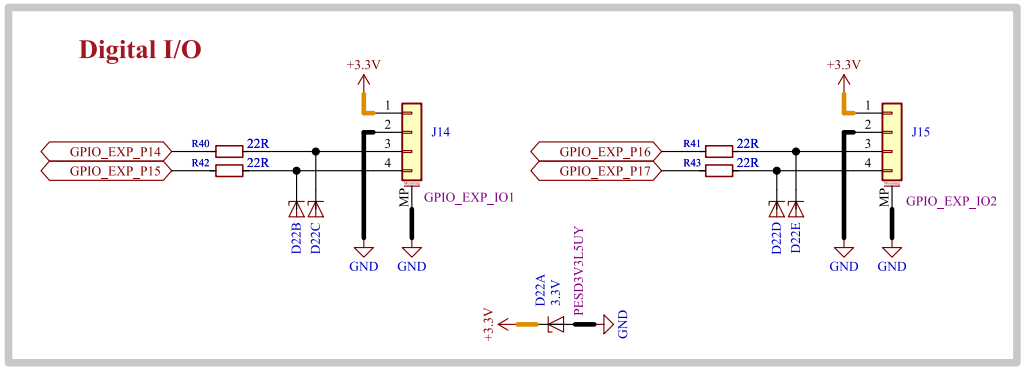
BUS

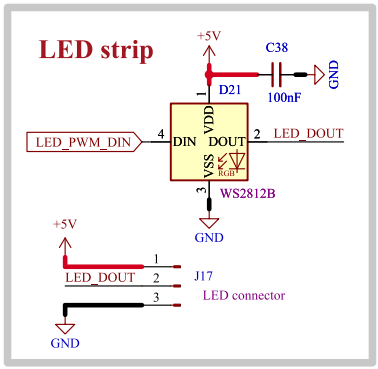


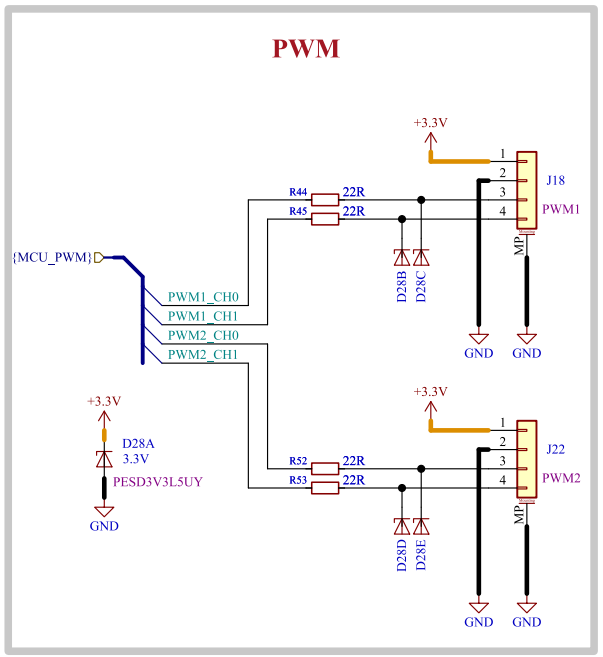
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	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Input&Output ADC	File Name: In&Out - ADC.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/In&Out - ADC/		Reviewer:	Size: **A4**	Sheet: **7** of **13**

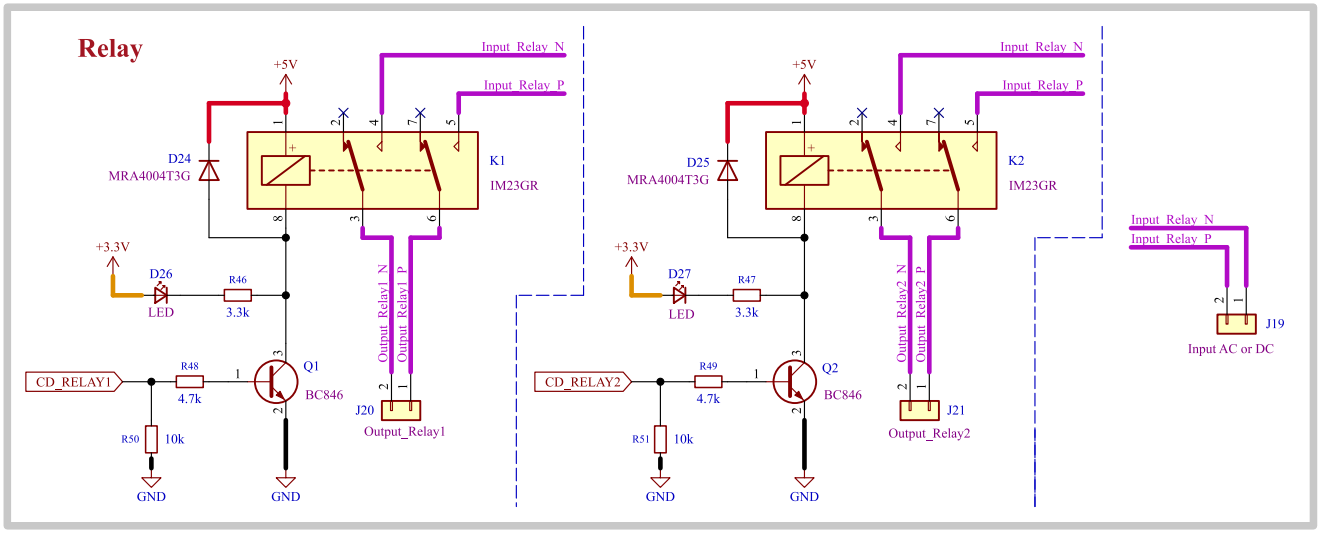
[8] Input&Output Command





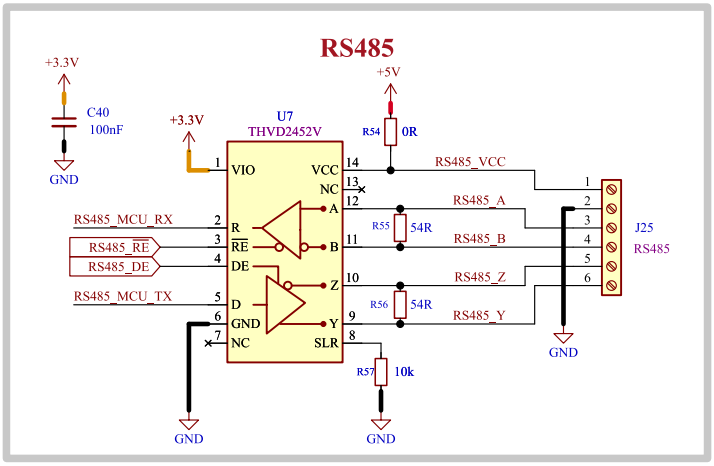


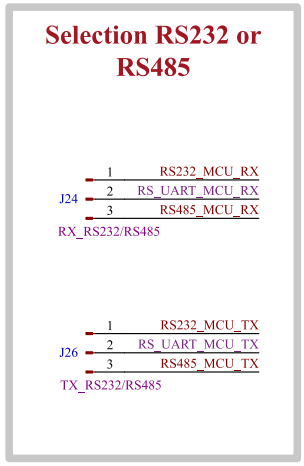


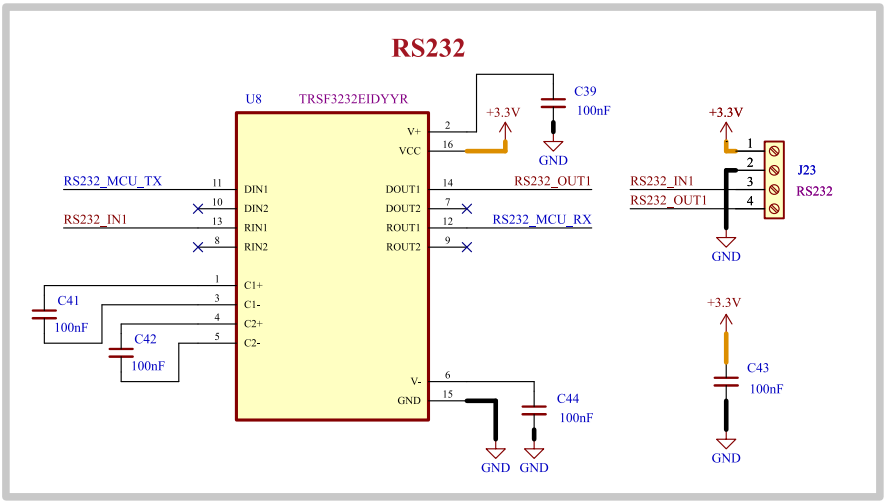


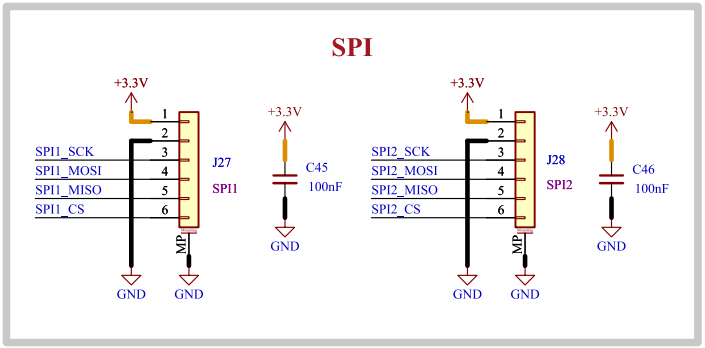
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 41b59e0
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Input&Output Command	File Name: In&Out - Command.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/In&Out - Command/		Reviewer:	Size: **A4**	Sheet: **8** of **13**

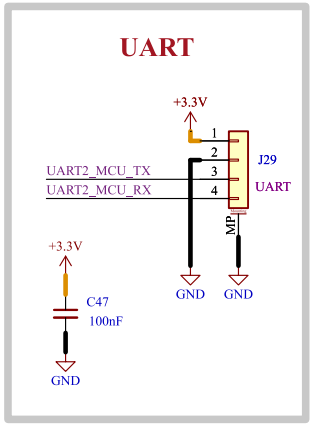
[9] Serial Interface

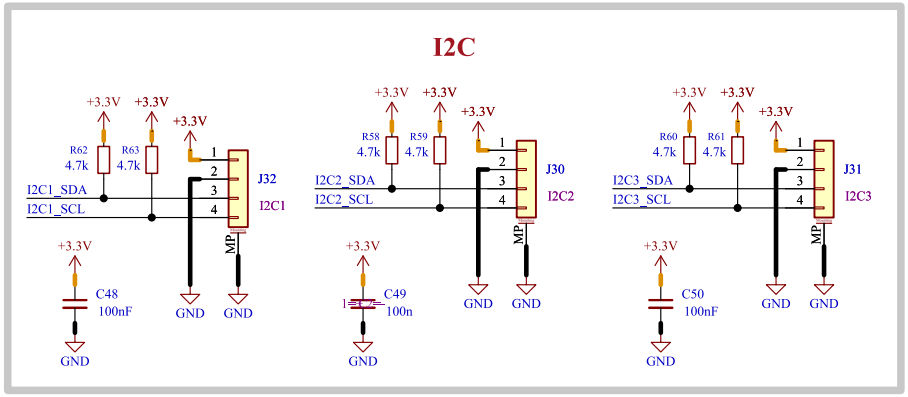


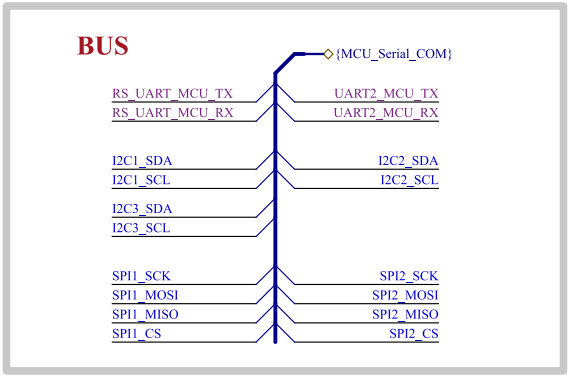








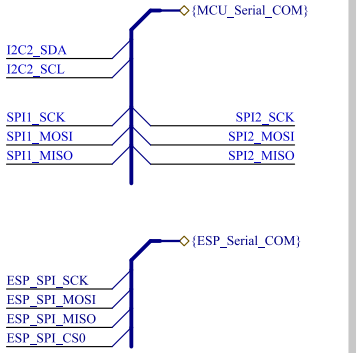




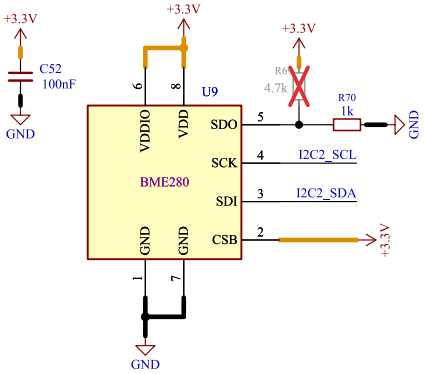
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 41b59e0
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Serial Interface	File Name: Serial Interface.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Serial Interface/		Reviewer:	Size: A4	Sheet: 9 of 13

[10] Sensor

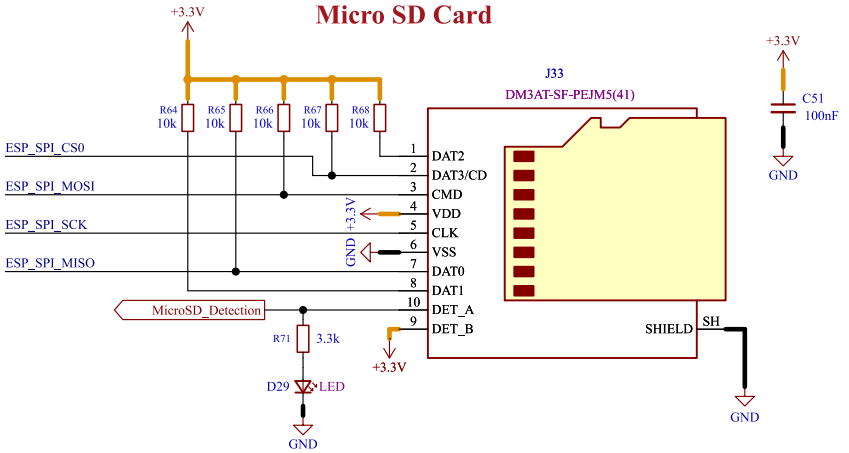
BUS



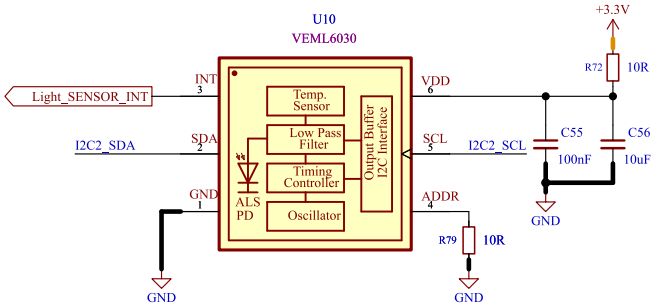
BME280 : Temperature, humidity and Pressure sensor



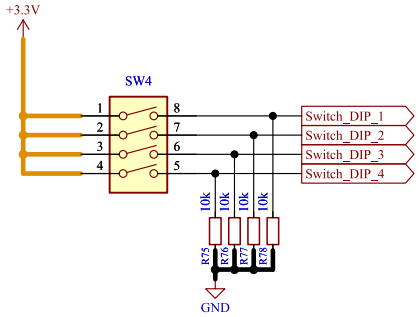
Micro SD Card



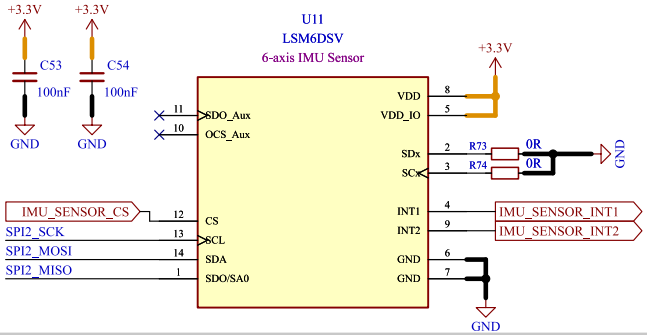
VEML6030 : Light sensor



Switch DIP



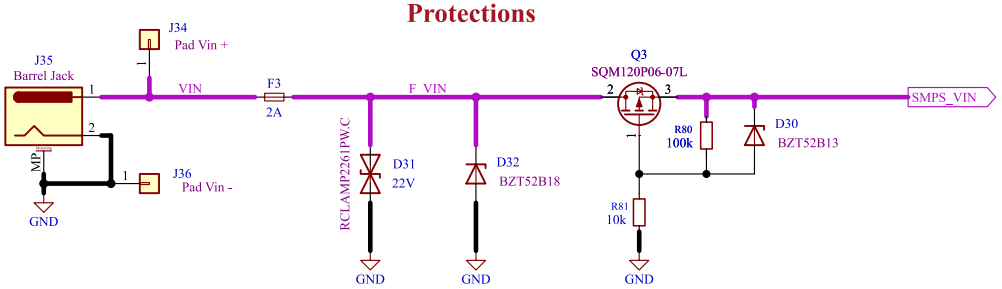
BME280 : IMU Sensor (Accelerometer and Gyroscope)



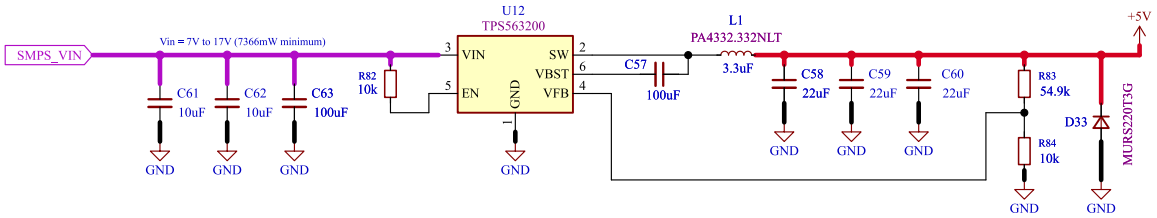
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 41b59e0
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Sensor	File Name: Sensor.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensor/		Reviewer:	Size: A4	Sheet: 10 of 13

[11] Power

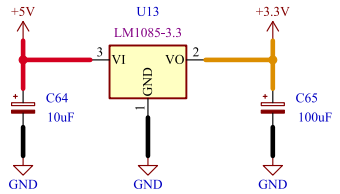
Protections



5V SMPS Regulator



3V3 LDO Regulator

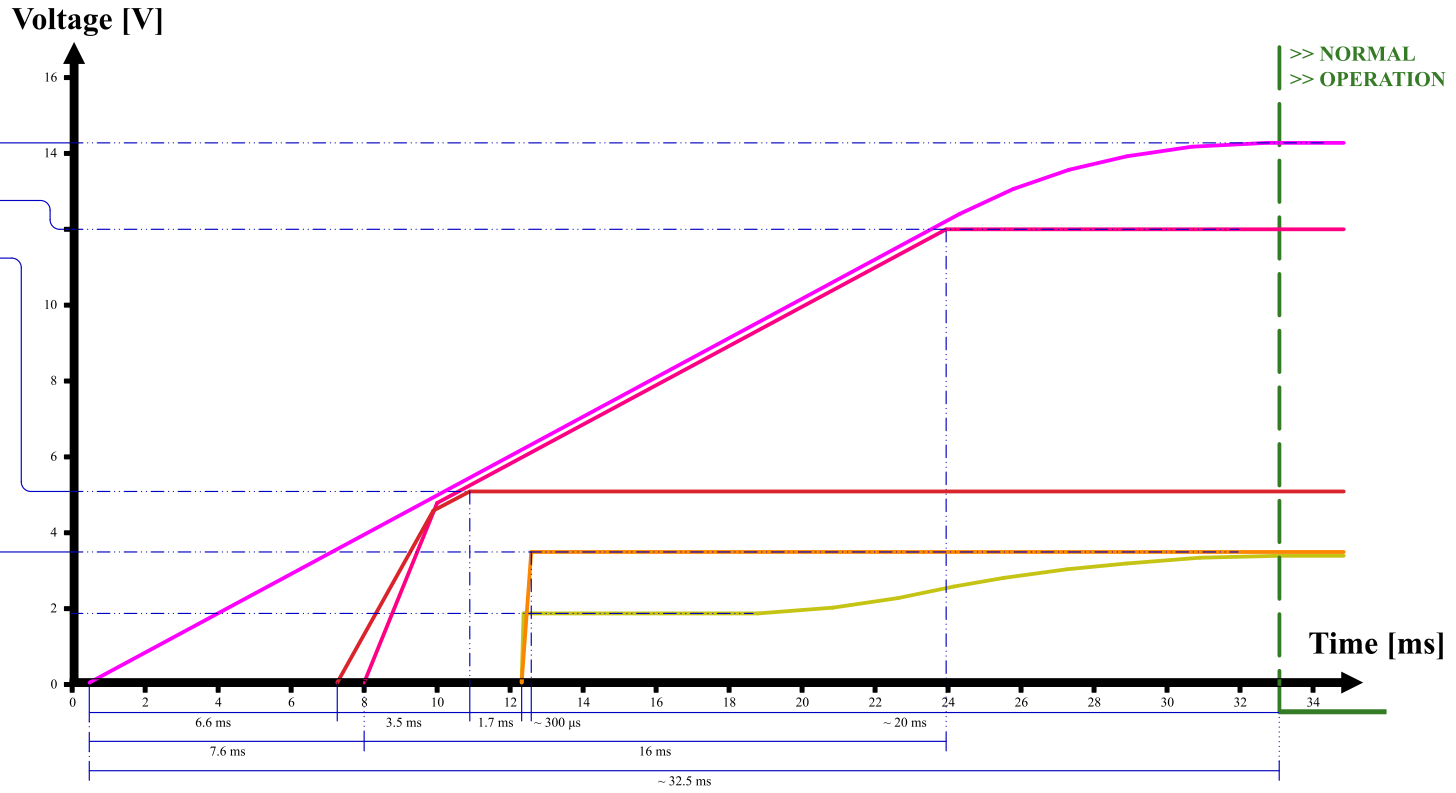


Protections:
->Overvoltage
->Voltage spike
->Overcurrent
->Reverse Polarity

Comments: LM1085-3.3 Datasheet; TPS563200 Datasheet	Company: **Nexvantis**		Variant: DRAFT	Git Hash: 41b59e0
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Power	File Name: Power.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power/		Reviewer:	Size: **A4**	Sheet: **11** of **13**

[12] Power - Sequencing

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 41b59e0
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 12 of 13

[13] Revision History

Comments:	Company: **Nexvantis**		Variant: DRAFT	Git Hash: 41b59e0
	Board Name: **DAQ Core**		Project Name: **Data Acquisition Board**	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: **A4**	Sheet: **13** of **13**